



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2010

INFORMATION TECHNOLOGY: PAPER I

Time: 3 hours

180 marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 16 pages and an Answer Sheet of 2 (i – ii) pages. Please check that your question paper is complete.
 2. Read the questions carefully and make sure that you answer all parts of each question.
 3. Answer ALL questions.
 4. Questions 3.3, 4.1, 6.5 and 7.6 must be answered on the Answer Sheet provided.
 5. Show all working where applicable.
 6. Non-programmable calculators may be used.
 7. Number your answers exactly as the sub-questions are numbered.
 8. Start your answer to each question on a new page.
 9. Please leave a line open between sub-questions.
 10. It is in your own interest to write legibly and to present your work clearly.
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KidBook is a social networking tool. It provides many social networking features to its members, including forums. In this paper we will look at many aspects of ICT through referring to the KidBook site.

The following is an excerpt from an online forum:

User	Topic: Introduction to the IT paper
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	October/November 2010 07h30 Hi guys Adam and I are registered users on the Kidbook site (www.kidbook.co.za). Kidbook is similar to Facebook, except that it caters specifically for school children and offers many educational aids.
 Adam Cape Town frequent_user Date joined: 24 May 2008	October/November 2010 07h32 Yeah, we've met the creators of Kidbook – they are Jordan, Selinah, Rajan and Christine. They've got this cool social network system supported by some great ICT kit.
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	October/November 2010 07h35 You'll be reading more about us later on, but first, in Question 1, you need to explain some computer terms to us. Thanks for the help ☺

QUESTION 1

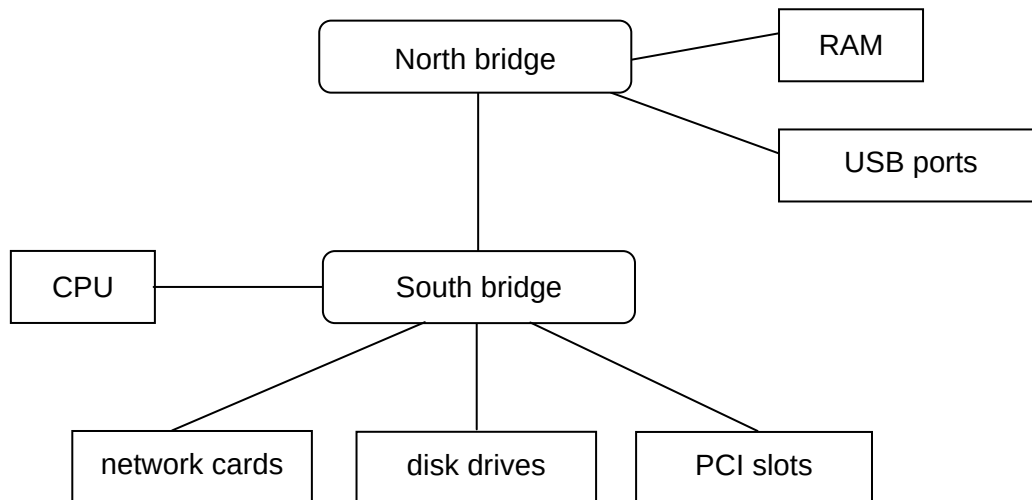
Briefly explain the following terms:

- 1.1 crosstalk
- 1.2 NTFS
- 1.3 Trojan horse
- 1.4 Android
- 1.5 keylogger
- 1.6 thin client
- 1.7 OSS
- 1.8 RFID
- 1.9 dual core CPU
- 1.10 Web 2.0

[20]

QUESTION 2

- 2.1 Jordan tried to explain to Selinah the difference between the North bridge and South bridge components on the motherboard. He drew the following diagram:



This diagram is not correct as some components are connected to the wrong bridge.

Redraw the diagram correctly. NB: Make sure you include all labels.

(4)

- 2.2 Different CPUs can have different instruction sets.

2.2.1 What is an instruction set?

(2)

2.2.2 Pipelining is a technology that allows a CPU to work on multiple instructions at once. Briefly describe how it does this.

(2)

2.2.3 Briefly describe and explain how ONE specific requirement for an efficient pipeline leads to most modern CPUs using a RISC style instruction set.

(2)

- 2.3 Even with different instruction sets, the basic task of the CPU remains the same. Name FOUR steps in the machine cycle of a CPU.

(4)

- 2.4 All CPUs contain certain basic components, one of which is a set of registers.

2.4.1 Give ONE example of what is stored in a register.

(1)

2.4.2 Name the other part of the CPU which is used for the temporary storage of a large amount of data and/or instructions.

(1)

- 2.5 Several technologies have been invented to improve the performance of a CPU. Explain how each of the following improves the speed of the CPU:

2.5.1 Reduced die size

(3)

2.5.2 Multi-threading

(2)

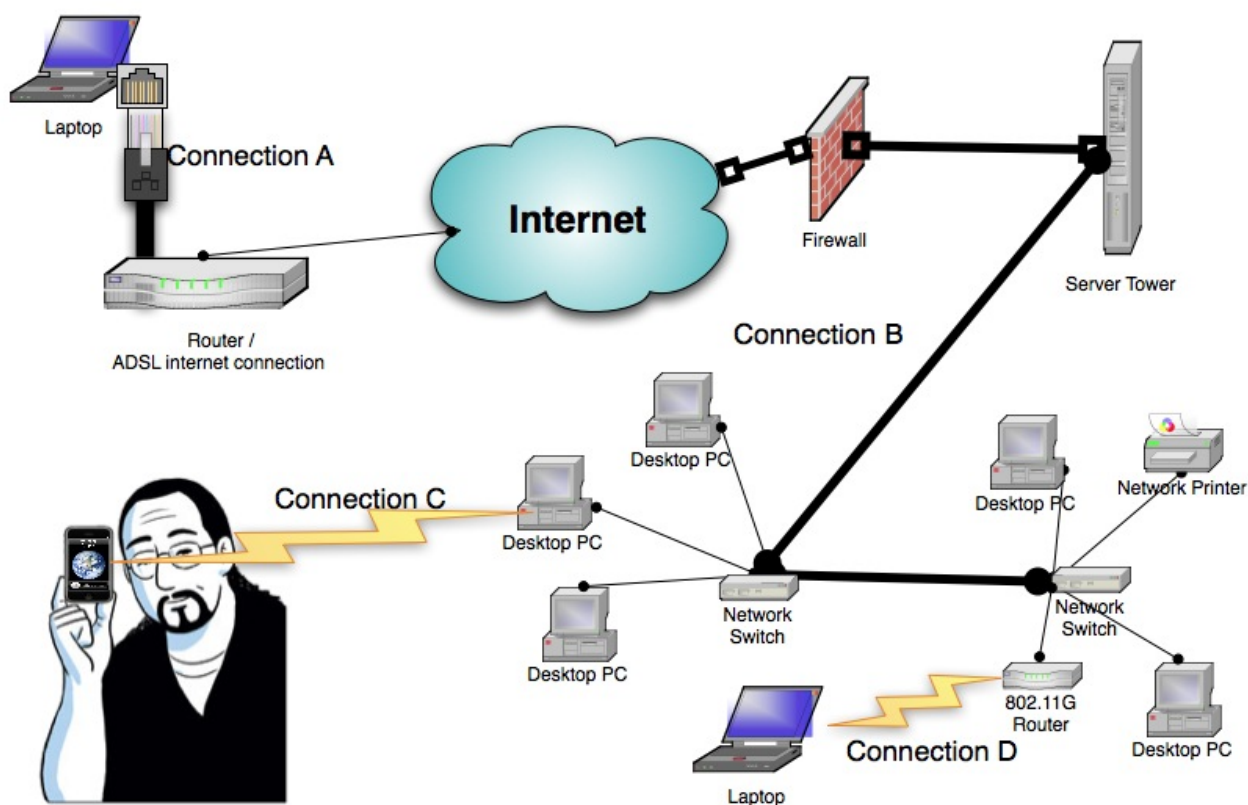
2.5.3 Multi-processing

(2)

[23]

QUESTION 3

- 3.1 In each of Kidbook's computers, the BIOS program is the first program that runs when a computer is started up. What are TWO of the things the BIOS does when the computer is switched on? (2)
- 3.2 In the KidBook offices there is a web server and a proxy server.
- 3.2.1 Describe what a proxy server does. (2)
- 3.2.2 Name TWO other kinds of servers besides a web server and a proxy server. (2)
- 3.3 The entire network is protected by anti-virus software, which is considered utility software. Give THREE other examples of utility programs and explain what each program does. NB: Answer on the Answer Sheet provided to you. (6)
- 3.4 How is it possible that Sibongile and Adam can print the same document on printers made by different companies and get identical results? Refer to drivers and the operating system in your answer. (2)
- 3.5 Rajan has been trying to convince the others that they should make use of cloud computing.
- 3.5.1 What is cloud computing? (2)
- 3.5.2 Explain why you would EITHER
- Trust and use cloud computing
 - OR
 - Refuse to use cloud computing.
- You must give a minimum of THREE clear reasons to support either argument. (3)
- [19]**

QUESTION 4 NETWORKING

- 4.1 The network diagram above shows a laptop connected to KidBook through the Internet, KidBook's internal network and Rajah connected to his desktop PC with his smartphone.

Take a careful look at all the labelled connections in the diagram. Each connection uses a specific type of technology (e.g. co-axial cable) and a physical medium (co-axial cable uses the physical medium of copper wire).

Complete the table on the Answer Sheet stating the BEST connection technology and the physical medium that technology uses for EACH of the labelled connections in the diagram.

(8)

- 4.2 Data travels from KidBook's server to Adam's computer via several routers.

4.2.1 What is the purpose of a router? (2)

4.2.2 How does a router differ from a switch? (2)

- 4.3 Sibongile is having trouble connecting to the network. She phones the helpdesk and they say she probably has a problem with the protocol she is using.

4.3.1 What set of protocols needs to be on any computer that is used to browse the Internet? (1)

4.3.2 Name TWO of the protocols that form part of this suite. (2)

- 4.4 A school with an old network that runs a very old version of Netware and uses the IPX/SPX protocol is having trouble connecting to KidBook (and, indeed the whole Internet). They want you to analyse the situation and
- explain what the problem is.
 - suggest a solution.
- (2)
- 4.5 Name a network control protocol (or media access method) that is used in:
- 4.5.1 a star topology (1)
- 4.5.2 a ring topology (1)
- 4.6 Explain how packet switching works. You may use diagrams to aid your answer. (4)
- [23]**

QUESTION 5 INTERNET

- 5.1 Sibongile (in Johannesburg) and Adam (in Cape Town) often communicate via Skype. Skype uses VoIP to operate.
- 5.1.1 What does VoIP stand for? (1)
- 5.1.2 What would motivate them to use VoIP instead of a cellphone? Give THREE reasons. (3)
- 5.2 The following post appears on Kidbook's IT forum:

User	Topic: Application software
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	10 September 2010, 14h25 Heya! See these terms below – my teacher has explained them all, but I still don't get why they are a threat/annoyance. Please help! virus hoax spam trojan horse phishing

Write an answer to this post by explaining what THREAT/DANGER/ANNOYANCE they present. NB: for each item you should present ONE fact per mark.

- 5.2.1 virus hoax (2)
- 5.2.2 spam (2)
- 5.2.3 trojan horse (2)
- 5.2.4 phishing (3)

- 5.3 Sibongile reads the Maths forum on Kidbook and sees a post with her name and picture. She is surprised to see this because she did not submit this post. Read the posts below and answer the questions that follow:

User	Topic: Factorising
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	12 September 2010, 16h35 Hi guys Can someone help me with this factorising squares business? I don't understand why it's wrong to say that $x\text{-squared} + y\text{-squared} = (x - y)(x + y)$. My teacher's tried to explain it, but I don't get it.
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	12 September 2010, 18h15 Hey, I did not submit this post! I'm not that stupid, I know you can't factorise $x\text{-squared} + y\text{-squared}$! That's my name and my photo on the post – what's going on?
 Adam Cape Town frequent_user Date joined: 24 May 2008	12 September 2010, 18h50 Oh, my goodness, you've been a victim of identity theft. Don't you have an anti-spyware program on your computer?

- 5.3.1 What is identity theft? (2)
- 5.3.2 Explain spyware by referring to what it can track. Give THREE examples. (3)
- 5.3.3 Explain how spyware can lead to identity theft. (2)
- 5.3.4 Apart from having an anti-spyware program, give TWO other actions or precautions that Sibongile should take to prevent further identity theft. (2)

5.4 Below is a thread regarding Maths help:

User	Topic: Maths help
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	13 September 2010, 15h15 Hey, everyone, I found this cool site to help me with Maths. It's called Maths Whizz. Just Google Maths Whizz, you'll find it.
 Adam Cape Town frequent_user Date joined: 24 May 2008	13 September 2010, 15h50 I had a look. How do you know the examples on the Maths Whizz site are all correct? Maybe some of the authors don't really know what they are doing?

Name THREE guidelines that Sibongile could use to judge whether to trust the Maths Whizz site. (3)

5.5 Maths Whizz offers 'Sample Examinations' for downloading, at R120 per examination paper, payable by credit card. The 'payment page' uses SSL to ensure that the payment is secure.

How does SSL:

- 5.5.1 keep the content that is being transferred private, i.e. make sure no one else can read it? (1)
- 5.5.2 allow your computer to check that it is really the 'Maths Whizz' computer that you are connected to? (1)

5.6 Sibongile submitted another post regarding a website that she found.

User	Topic: Software
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	16 September 2010, 15h05 Hi guys, I need information on application software, and found some stuff (pasted below) on the ITforKids website. Well, this author only presents four categories of application programs. There are many more. Also, the definitions are not very good, and the information is poorly presented. I need a better source of information on application software – can someone please help me with this? Below is what I found on the ITforKids website, followed by the URL for that website. ----- Categories of application software: Spreadsheets are programs that can be used to do calculations. Presentation programs are used to put together slide shows for content that you want to present to a group of people. Database programs are used to store information, such as the details of the employees in your company. Graphics programs are used to create or edit photographs. <http://www.ITforkids.com/software/applicationSoftware/categories.html>

When writing this post, Sibongile copied and pasted the information from the **ITforKids** site (everything below the dotted line). Give TWO reasons why she has not violated copyright in this post.

(2)
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QUESTION 6 SOCIAL AND ETHICAL ISSUES

- 6.1 The KidBook site offers educational resources to its members, but it is also a social networking site, similar to Facebook. The data that people make available on these types of sites can put them at risk. Give THREE possible reasons why businesses and/or schools dislike these sites and sometimes ban them. (3)
- 6.2 Below is an excerpt from KidBook's Privacy Statement. Study the statement and answer the questions that follow.

Introduction

This Privacy Policy deals with personally identifiable information that may be collected by us on our site.

1. Collection of Data

Our registration process requires only a valid email address and a unique user ID and password. Providing us with other information is optional. Please take note that your user name, email address, or other submissions that you make on this site may contain your real name and may appear on this site. We may also automatically receive general information such as your IP address and cookie information.

2. Use of Data

We will make efforts so that your data will not be provided to third parties unless we obtain your consent.

3. Cookies

Advertisements may appear on our website and, if so, may set and access cookies on your computer and is subject to the privacy policy of those parties providing the advertisement. However, the companies serving the advertising do not have access to our cookies.

[Adapted from: <<http://forums.digitalpoint.com>>]

- 6.2.1 Why is it necessary for the KidBook site to have a privacy statement? (2)
- 6.2.2 How does a cookie improve the browsing experience for the user? (3)
- 6.2.3 Does the existence of cookies pose a threat to a computer user? (2)
- 6.3 Kidbook Company decides to start developing a Wiki.
- 6.3.1 How does a Wiki differ from other reference sources? (2)
- 6.3.2 What are the problems related to the content found in a Wiki? (2)
- 6.4 Christine works for KidBook. She suddenly resigns and announces plans to start her own business. Jordan is Kidbook Company's network administrator. He is responsible for ensuring that the data on the network is secure and for managing users. Give TWO actions that Jordan must take when Christine leaves KidBook so as to maintain the company security. (2)

6.5 Below is a list of computer related careers, followed by excerpts from Sibongile's and Adam's Kidbook profiles.

Study the profiles carefully and choose:

6.5.1 TWO possible computer careers for Sibongile.

6.5.2 ONE possible computer career for Adam.

Answer this question on the Answer Sheet. **Explain** your choices carefully by referring **directly** to facts or sentences from these learners' profiles. (9)

List of computer careers:

- network technician
- network administrator
- IT teacher
- sole owner of a business which sells PCs
- individual programmer (working from home, delivering code to customers via email)
- team programmer (part of a large programming team, working under a leader)



Sibongile

Subjects

English 75
Afrikaans 78
Maths 85
Visual Arts 73
Business Studies 74
Accountancy 85
IT 71
LO 68

Sports and clubs

First team netball
Teaches Bible class at church on Sundays
Journalism club (chairman)

Achievements

Chairperson of Matric Dance committee

Personal introduction:

I have a wide range of interests – I enjoy cooking and painting, but I can also spend hours at the computer; I love programming. Maybe I'll choose a career in IT, or maybe I'll become a journalist, or I might choose to work in a restaurant.



Adam

Subjects

English 76
Afrikaans 76
Maths 83
Physical Sciences 81
Life Sciences 80
IT 92
LO 71

Sports and clubs

First team swimming
Chess club
Photography club
Soccer

Achievements

High score in Computer Olympiad
Photography award by *SuperPhoto* magazine

Personal introduction:

I love solving problems, that's why I enjoy chess. My big hobby is photography. I spend a lot of time on my own in the bush to find good shots of landscapes; I also do a lot of night time photography – I like to explore the technical side of my camera.

6.6 Here is another post by Sibongile:

User	Topic: Careers
 Sibongile Johannesburg frequent_user Date joined: 11 Feb 2009	14 September 2010, 14h15 I admire Gordon Ramsay and Jamie Oliver, but I don't want to be a public chef. I want to make a living writing and publishing recipe books.
 Adam Cape Town frequent_user Date joined: 24 May 2008	15 September 2010, 18h50 I think writing recipe books will be a waste of time. Nowadays people can look up recipes on the Internet, they don't have to buy recipe books.

Does the existence of the Internet mean that Sibongile should abandon her plan to be an author of recipe books, or does the Internet offer an advantage to her? Give THREE advantages and/or disadvantages of the Internet in this situation. (3)

6.7 As part of a fund-raising event, Sibongile's class holds a 48-hour cricket game. Players in the two teams play for as long as they can, and replacement players can wait at the field or at home, from where they will be called if needed. Give TWO advantages of using Twitter during this event. (2)

6.8 The issue of global warming is currently very serious. Imagine that the whole world stopped using computers tomorrow.

6.8.1 How would this contribute to the reduction of global warming? (1)

6.8.2 Give TWO reasons why having no computers would make finding solutions to global warming more difficult. (2)

[33]

QUESTION 7 DATABASE NORMALISATION, SQL AND PROGRAMMING

7.1 Sibongile created the following table about her friends' hobbies:

tblHobbies(name, age, hobby)

name	age	hobby
Thabo	16	Photography
Thabo	16	Mountaineering
Mary	15	Hip hop
Mary	15	Photography
Mary	15	Guitar
Ayesha	18	Guitar
Adam	17	Ice skating
Adam	17	Ballet

Explain why this table has redundant data. (2)

7.2 What is meant when saying a table has a composite key? (1)

7.3 Consider the following situation:

KidBook's offices are in a building where the parking garage is protected by security measures. Every time a person enters the parking garage, the guard on duty records the following:

- driver's ID number
- car registration number
- timestamp (date and time of entry)
- number of passengers
- purpose of visit

This information is stored in a database table called **tblParking** with the fields (**driverID**, **carReg**, **timestamp**, **passengers**, and **purpose**).

7.3.1 Explain why **driverID** is not suitable to be chosen as a primary key for this table. (1)

7.3.2 If you had to create a composite key for this table, which fields would you use? (3)

- 7.4 The following two tables show some KidBook users as well as donations received for the KidBook Educational Fund.

tblUsers(username, status, emailAddress)

username	status	emailAddress	posts
digitalgal	user	digitalgal@gmail.com	12
skatingdude	frequent_user	skatingdude@yahoo.com	58

tblDonations(donationID, amount, timestamp, username)

donationID	amount	timestamp	Username
424	R500	22 Jan 2010 11:36	Digitalgal
425	R250	26 Jan 2010 15:20	Skatingdude
426	R650	23 Feb 2010 07:30	Digitalgal
427	R320	18 Mar 2010 10:24	Skatingdude
428	R1 500	4 May 2010 12:25	Skatingdude

Explain why the following SQL statement:

DELETE from tblUsers WHERE username = 'digitalgal'
would corrupt the data in the database.

(2)

- 7.5 A SQL query was created to show:

- the username
- the number of donations per user
- the total received in donations by each user

The results of the query are shown below. In the example below the total of digitalgal's two donations comes to R1 150, and the total of skatingdude's three donations comes to R2 070.

username	donations	total
digitalgal	2	1 150
skatingdude	3	2 070

Apart from the keywords 'SELECT' and 'FROM', name all the keywords and functions that were used to create this query.

(8)

- 7.6 Kidbook Company needs a program which will be used to maintain the users' table: tblUsers.

tblUsers

username	status	emailAddress	posts
digitalgal	user	digitalgal@gmail.com	12
skatingdude	frequent_user	skatingdude@yahoo.com	58

The program will make use of an object called **User** which will have the following capabilities:

- Increase a user's posts by one when called to do so.
- Checking a user's current number of posts and upgrading the user's status if applicable.

Complete the table on the supplied Answer Sheet to design this class.

NB: You do not need to plan a constructor or Get/Set methods.

(6)

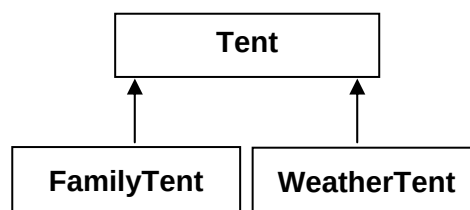
- 7.7 Adam is writing a program for an outdoor shop. The shop sells the following kinds of tents:

- **Tent** a basic tent
- **FamilyTent** comes with a gazebo (additional covered area, roof only, no sides)
- **WeatherTent** the same as FamilyTent, but has clip-on sides for the gazebo (to make it an enclosed 'room') – there can be two, three or four clip-on sides

The program needs to work with the following attributes for each of these types of tents:

Tent floorArea shape height	FamilyTent floorArea shape height gazeboArea	WeatherTent floorArea shape height gazeboArea gazeboSides
---	---	---

Adam plans to have three classes: Tent, FamilyTent and WeatherTent. Below is his rough sketch for a class hierarchy:



- 7.7.1 Explain why Adam's design does not make maximum use of the advantages offered by inheritance.

(3)

- 7.7.2 Draw an improved hierarchy diagram.

(2)

- 7.8 What is the purpose of an access modifier? (2)
- 7.9 Give one example of an access modifier. (1)
- 7.10 When would you choose to write a ***typed*** method instead of an ***untyped*** method? (2)
- [33]

180 marks

Total: 180 marks